CROPGRO Species Coefficients (Simple Definitions 2024) – adapted for Soybean

Notes:

- Parameter descriptions in black: descriptions listed in the 2006 Simple Definitions file
- Parameter description in Green: descriptions found in the code.
    - Routine/sub-routine where the descriptions are in parentheses and underlined.
- Blue: Ken Boote's comments.
- Yellow: functions with curves below the table.
- Purple: to review (if the info is correct, placed in the proper section, or needs to be removed).

!*PHOTOSYNTHESIS PARAMETERS (canopy photosynthesis) – to be discontinued

Table 1

Parameter	Parameter description	Soybean
PARMAX [1]	Value of PAR at which photosynthesis is 63% of the maximum value (PHTMAX). Used in daily canopy photosynthesis calculations (moles[quanta]/m2-d). (/Plant/CROPGRO/PHOTO.for) Moles of PPFD at which assimilate is 63% of maximum	40.0
PHTMAX [1]	Maximum amount of CH20 which can be produced if photosynthetically active radiation (PAR) is very high (3 times PARMAX) and all other factors are optimal (g[CH2O]/m2-d). (/Plant/CROPGRO/PHOTO.for) The asymptote (g CH2O/m2/day)	61.0
KCAN	Canopy light extinction coefficient for daily PAR, for equidistant plant spacing, modified when in-row and between row spacing are not equal. (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/CANOPY.for ; /Plant/CROPGRO/IPPLNT.for not listed at the bottom ; /Plant/CROPGRO/PHOTO.for ; /Plant/CROPGRO/SENES.for /SPAM/TRANS.for ; /Plant/CROPGRO/VEGGR.for not listed at the bottom) Light extinction coefficient for daily canopy option, dimensionless	0.68
CCMP [2]	Canopy CO2 compensation point (CO2 at which daily PG is 0.0). (/Plant/CROPGRO/PHOTO.for) canopy compensation point	80.0
CCEFF [2]	Relative efficiency of CO2 assimilation used in equation to adjust canopy photosynthesis with CO2 concentrations. (/Plant/CROPGRO/PHOTO.for) initial slope	0.0105
CCMAX [2]	Maximum daily canopy photosynthesis relative to photosynthesis at a CO2 concentration of 330 vpm. (/Plant/CROPGRO/PHOTO.for) normalized point that indicates that canopy rate can be 2.09 times greater at saturating CO_2 than at 350 ppm CO_2	2.09
FNPGN(4), TYPPGN (Fig.1)	FNPGN(I) Critical leaf N concentration for function to reduce photosynthesis due to low leaf N levels (4 values for function CURV). TYPPGN Type of function for the leaf N effects on PG. (/SPAM/ETPHOT.for, not listed at the bottom ; /SPAM/ETPHR.for, not listed at the bottom ; /Plant/CROPGRO/PHOTO.for)	1.90 5.50 20.0 20.0 QDR

	a two-sided generic curve, in this case, using only the two leftmost points to describe both canopy and leaf photosynthesis response to leaf N concentration	
FNPGT(4), TYPPGT [3] (Fig.2)	FNPGT(I) Critical values of temperature for the functions to reduce canopy PG under non-optimal temperatures (in function CURV). TYPPGT Character variable specifying the type of function to use for the relationship between temperature and PG (for use in function subroutine CURV). (/Plant/CROPGRO/PHOTO.for) a two-sided generic curve, in this case, LIN means linear 4-point lookup, that describes daily canopy assimilation in response to average daytime temperature	3.00 22.0 34.0 45.0 LIN
XPGSLW(1-10) YPGSLW(1-10)	XPGSLW(I): Array of SLW values for table look-up function, used with YPGSLW (g[leaf] / m2[leaf]). YPGSLW(I) Array of PG values corresponding to SLW values in array XPGSLW (g[CH2O] / m2 / d). (/Plant/CROPGRO/PHOTO.for) Relative change in DAILY canopy assimilation with change in average canopy SLW, normalized to 1.00 at SLWREF, which is where leaf photosynthesis is defined.	0.0 .001 .002 .003 .0035 .004 .005 .006 .008 .010 .162 .679 .867 .966 1.000 1.027 1.069 1.100 1.141 1.167
PGREF	Reference value for leaf level photosynthesis used in canopy light response curve (µmol[CO2] / m2-s). (/Plant/CROPGRO/PHOTO.for) Used only for the "Daily" canopy assimilation option. Allows cultivar variation in LFMAX to change daily canopy assimilation. PGREF should really be in the middle of, or represent the standard cultivars used, such as Bragg. Not used for the leaf-level, hedgerow photosynthesis option.	1.030

[1] PARMAX and PHTMAX - define the asymptotic response of daily canopy assimilation to daily photosynthetic photon flux. Increase PHTMAX to increase dry matter growth for daily option only.

[2] CCMP, CCMAX, and CCEFF - All normalized, define the response of daily canopy assimilation to CO_2, for daily version.

[3] FNPGN(4) and TYPPGN - This is a two-sided generic curve, in this case, using only the two leftmost points to describe both canopy and leaf photosynthesis response to leaf N concentration. QDR defines a quadratic curve as photosynthesis increases from zero at the minimum leaf N concentration to its maximum at the maximum leaf N concentration. Same leaf-derived function is used for leaf and canopy options; however, subroutine PHOTO modifies it for DAILY option with a square root change.

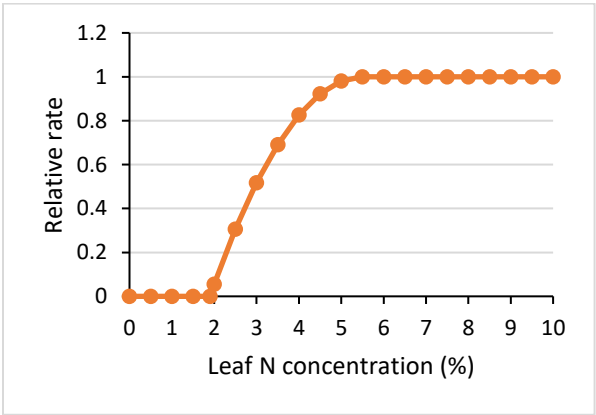

Figure 1. Relative rate of canopy and leaf photosynthesis response to leaf N concentration

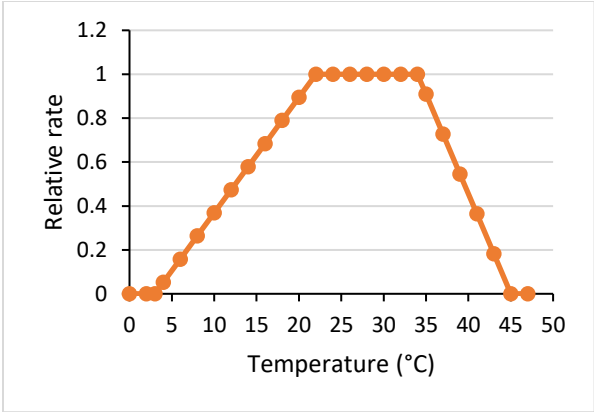

Figure 2. Relative rate of daily canopy assimilation in response to average daytime temperature.

!*PHOTOSYNTHESIS PARAMETERS (leaf photosynthesis)

Table 1.1

Parameter	Parameter description	Soybean
FNPGN(4), TYPPGN (Fig.1)	FNPGN(I) Critical leaf N concentration for function to reduce photosynthesis due to low leaf N levels (4 values for function CURV). TYPPGN Type of function for the leaf N effects on PG (/SPAM/ETPHOT.for, not listed at the bottom ; /SPAM/ETPHR.for, not listed at the bottom ; /Plant/CROPGRO/PHOTO.for) a two-sided generic curve, in this case, using only the two leftmost points to describe both canopy and leaf photosynthesis response to leaf N concentration	1.90 5.50 20.0 20.0 QDR
XLMAXT (Fig.3) YLMAXT (Fig.3)	Temperature effect on MAXIMUM LEAF PHOTOSYNTHESIS (LMXREF) at saturating CO2 via lookup table. Sawtooth shape i.e. linear increase to peak, then linear decrease. Based on analysis of LMXREF using Tenhunen's (1976) data with QEFF from Ehleringer and Bjorkman (1977). TEMPMX (Max.temperature?) scaled to 1.0 at 30 oC. (/SPAM/ETPHOT.for, not listed at the bottom ; /SPAM/ETPHR.for, not listed at the bottom) a 6-point lookup function, that describes relative rate of photosynthetic electron-transport (YLMAXT) in response to temperature (XLMAXT). Used only for Leaf option and affects computed light-saturated leaf photosynthesis.	0.0 8.0 40.0 44.0 48.0 55.0 0.0 0.0 1.0 0.8 0.0 0.0
FNPGL(4), TYPPGL (Fig.4)	Minimum night temp effect on Pmax next day, quadratic function after Mike Bell, peanut. ONLY the first two numbers are used. KJB, 9/7/94. OKAY TO USE NIGHT MINIMUM CANOPY TEMPERATURE LATER. TYPPGL ?? (/SPAM/ETPHR.for, not listed at the bottom) Describes relative effect of minimum night temperature (TMIN) on next day's single leaf light-saturated photosynthesis rate. Use only the two leftmost values, assuming QDR quadratic/parabolic response from 0.0 to 1.0 (no effect) on leaf photosynthesis.	0.00 19.00 50.0 60.0 QDR
PGEFF (not found in the code) Removed or changed to QEREF?	Quantum efficiency of leaf photosynthesis, defined at 350 ppm CO2, 21% oxygen, and 30 C. Set from published values in literature, Ehlringer and Bjorkman.	.0541

SCV (not found in the code)	Scattering of photosynthetic photon flux by single leaves (transmittance and reflectance)	0.20
KDIF (not found in the code)	Diffuse beam extinction coefficient	0.80
LFANGB	(/SPAM/ETPHOT.for, not listed at the bottom) Leaf angle distribution (type), 2 = spherical, from Campbell and Norman, 1989. Plant canopies: their growth, form and function.	2.0
SLWREF	(/SPAM/ETPHOT.for, not listed at the bottom ; /SPAM/ETPHR.for, not listed at the bottom). Specific leaf weight at which LFMAX is defined	.0035
SLWSLO	(/SPAM/ETPHOT.for, not listed at the bottom ; /SPAM/ETPHR.for, not listed at the bottom). Rate of change in SLW with cumulative LAI depth	.0004
NSLOPE	(/SPAM/ETPHOT.for, not listed at the bottom ; /SPAM/ETPHR.for, not listed at the bottom). Rate of change in leaf N concentration with cumulative LAI depth	.3000
LNREF	LNREF Value of leaf N above which canopy PG is maximum (for standard cultivar). (/Plant/CROPGRO/PHOTO.for ; /SPAM/ETPHOT.for, not listed at the bottom ; /SPAM/ETPHR.for, not listed at the bottom). Leaf N concentration at which LFMAX is defined, it really should be at the maximum N concentration, but is a bit below.	4.90

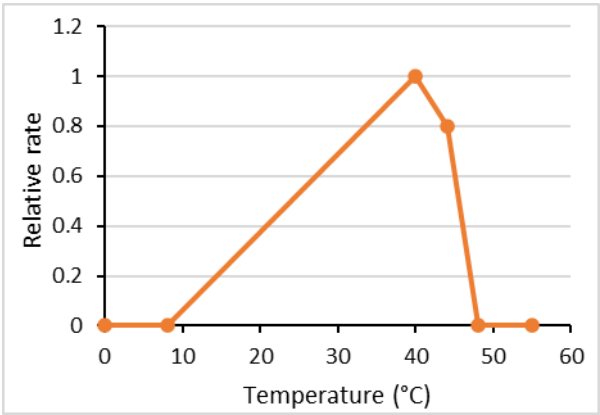

Figure 3. Relative rate of photosynthetic electron-transport (YLMAXT) in response to temperature (XLMAXT).

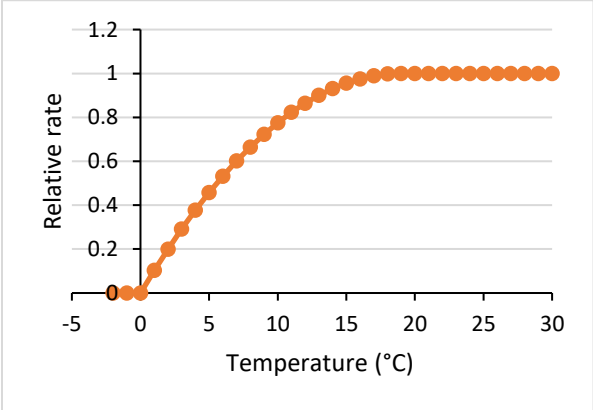

Figure 4. Relative rate effect (where 1.0 means no effect) of minimum night temperature (TMIN) on next day's single leaf light-saturated photosynthesis rate.

!*RESPIRATION PARAMETERS

Table 2

Parameter	Parameter description	Soybean
RES30C	Respiration coefficient that depends on gross photosynthesis, value at 30C (g CH2O/g DW/hr). (/Plant/CROPGRO/CROPGRO.for) Constant, that describes maintenance respiration as a function of total crop dry weight (minus oil, protein, and starch in the seed). Constant is temp. dependent.	3.5E-04 (0.0004) g CH_2O/g DW/hr
R30C2	Respiration coefficient that depends on total plant mass, value at 30C (g[CH2O] used / g[CH2O] fixed / hr). (/Plant/CROPGRO/CROPGRO.for). Constant, that describes maintenance respiration as a function of canopy photosynthesis. Constant is temperature dependent & summed over 24 hr.	.0040 g CH_2O/g CH_2O photosynthate/hr
RNO3C	Respiration required for reducing NO3 to protein (g[CH2O] / g[protein]). (/Plant/CROPGRO/CROPGRO.for). g of glucose required for growth respiration and synthesis of a gram of protein with nitrate as the N source.	2.556
RNH4C	CH2O required for protein synthesis when source of N is ammonium uptake (g[CH2O] / g[protein]). (/Plant/CROPGRO/CROPGRO.for). g of glucose required for growth respiration and synthesis of a gram of protein with ammonium as the N source. Same as nitrate because most N is converted to NO3 anyway, and there is uncertain added C costs for organic acid synthesis for cation balance for NH4-fed plants.	2.556
RPRO	Respiration required for re-synthesizing protein from mobilized (g[CH2O] / g[protein]). (/Plant/CROPGRO/CROPGRO.for). Cost to re-synthesize protein from reduced-N amino acids after breakdown of protein. Is this correct?	.360
RFIXN	CH2O required for biological N fixation (g[CH2O] / g[protein]). (/Plant/CROPGRO/CROPGRO.for). g of glucose required for growth respiration and synthesis of a gram of protein with N-fixation (hydrogenase-negative) as the N source.	2.830
RCH2O	Respiration required for synthesizing CH2O structure (g[CH2O] / g[tissue]). (/Plant/CROPGRO/CROPGRO.for)	1.242
RLIP	Respiration required for synthesizing lipid structure (g[CH2O] / g[lipid]). (/Plant/CROPGRO/CROPGRO.for)	3.106
RLIG	Respiration required for synthesizing lignin structure (g[CH2O] / g[lignin]). (/Plant/CROPGRO/CROPGRO.for)	2.174
ROA	Respiration required for synthesizing organic acids (g[CH2O] / g[product]). (/Plant/CROPGRO/CROPGRO.for)	.929
RMIN	Respiration required for synthesizing mineral structure (g[CH2O] / g[mineral]). (/Plant/CROPGRO/CROPGRO.for)	0.05
PCH2O	Respiration loss due to storage/mobilization of CH2O (g[CH2O] / g[CH2O]). (/Plant/CROPGRO/CROPGRO.for)	1.13

!*PLANT COMPOSITION VALUES

Table 3

Parameter	Parameter description	Soybean
PROLFI	Maximum protein composition in leaves during growth with luxurious supply of N (g[protein] / g[leaf tissue]) (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/VEGGR.for) "Maximum" protein concentration of leaf tissue	.356
PROLFG	Normal growth protein composition in leaves during growth (g [protein] / g [leaf tissue]). (/Plant/CROPGRO/VEGGR.for) "normal growth" protein concentration of leaf tissue	.285
PROLFF	Minimum leaf protein composition after N mining (g [protein] / g [leaf]). (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/PODS.for) "final" protein concentration of leaf tissue	.125
PROSTI	Maximum protein composition in stems during growth with luxurious supply of N (g[protein] / g[stem]). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/INCOMP.for ; /Plant/CROPGRO/IPPLNP.for ; /Plant/CROPGRO/VEGGR.for) "Maximum" protein concentration of stem tissue	.165
PROSTG	Normal growth protein composition in stems during growth (g [protein] / g [stem]). (/Plant/CROPGRO/VEGGR.for) "normal growth" protein concentration of stem tissue	.110
PROSTF	Minimum stem protein composition after N mining (g [protein] / g [stem]). (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/INCOMP.for) "final" protein concentration of stem tissue	.035
PRORTI	Maximum protein composition in roots during growth with luxurious supply of N (g[protein] / g[root]). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/INCOMP.for ; /Plant/CROPGRO/IPPLNT.for ; /Plant/CROPGRO/VEGGR.for) "Maximum" protein concentration of root tissue	.092
PRORTG	Normal growth protein composition in roots during growth (g[protein] / g[root]). (/Plant/CROPGRO/VEGGR.for) "normal growth" protein concentration of root tissue	.064
PRORTF	Minimum root protein composition after N mining (g[protein] / g[root]). (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/GROW.for) "final" protein concentration of root tissue	.056
PROSHI	Maximum protein composition in shells during growth with luxurious supply of N (g[protein] / g[shell tissue]). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/IPPLNT.for ; /Plant/CROPGRO/PODS.for)	.196

	Maximum protein composition in shells during growth with luxurious supply of N (g[protein] / g[shell tissue])","PODS, IPPLNT, I. (/Plant/CROPGRO/INCOMP.for)	
	"Maximum" protein concentration of shell tissue	
PROSHG	"normal growth" protein concentration of shell tissue	.196
PROSHF	Minimum shell protein composition after N mining (g[protein] / g[shell]) (/Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/PODS.for)	.056
	"final" protein concentration of shell tissue	
SDPROS	Seed protein fraction of standard cultivar at 25°C (g[protein] / g[seed])","INCOMP. (/Plant/CROPGRO/INCOMP.for)	.400
	Standard protein concentration of seed. Note: now only standard reference, changed by the cultivar-specific value in the ECO file. Also modified by C and N stresses as shown below.	
SDPROG	Standard "growth" protein concentration of seed. Note: now only standard reference, changed by the cultivar-specific value in the ECO file. Also modified by C and N stresses as shown below.	.400
PRONOD	Protein composition in nodules (g[protein] / g[nodule]). (/Plant/CROPGRO/GROW.for ; CROPGRO/NFIX.for)	.300
	Protein concentration of nodules	
PROMIN	Minimum N concentration of seed (g[N] / g[seed]). (/Plant/CROPGRO/PODS.for)	0.03
PROMAX	Maximum N concentration of seed (g[N] / g[seed]). (/Plant/CROPGRO/PODS.for)	0.08
THETA	Curvature of rectangular hyperbola to limit seed growth rate to hold minimum seed N concentration. (/Plant/CROPGRO/PODS.for)	0.8
	Curvature of rectangular hyperbola to limit (reduce) seed growth rate to hold minimum seed N concentration. Value of zero is Michaelis-Menten and 1.0 is a Blackman two-line limiting (latter allows seed to grow at normal rate, but lower N concentration until the absolute limit (PROMIN) is reached). Curvature of 0.7 to 0.8 works best to slowly limit seed growth at initially declining seed N concentration.	
PCARLF	Proportion of leaf tissue that is carbohydrate (fraction). (/Plant/CROPGRO/GROW.for)	.405
	Proportion of leaf tissue that is carbohydrate (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	
PCARST	Proportion of stem tissue that is carbohydrate (fraction) (/Plant/CROPGRO/GROW.for)	.649
	Proportion of stem tissue that is carbohydrate (fraction)","IPGROW, INCOMP (/Plant/CROPGRO/INCOMP.for)	
PCARRT	Proportion of root tissue that is carbohydrate (fraction). (/Plant/CROPGRO/GROW.for)	.711
	Proportion of root tissue that is carbohydrate (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	
PCARSH	Proportion of shell tissue that is carbohydrate (fraction). (Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/IPPLNT.for)	.434
	Proportion of shell tissue that is carbohydrate (fraction)","IPGROW, INCOMP, IPPLNT. (/Plant/CROPGRO/INCOMP.for)	
PCARSD	Proportion of seed tissue that is carbohydrate (g[CH2O] / g[seed]).	.315

	(/Plant/CROPGRO/GROW.for) Proportion of seed tissue that is carbohydrate (g[CH2O] / g[seed])","IPGROW, INCOMP, IPPLNT. (/Plant/CROPGRO/INCOMP.for)	
PCARNO	Proportion of nodule tissue that is carbohydrate (fraction). (/Plant/CROPGRO/GROW.for) Proportion of nodule tissue that is carbohydrate (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.480
PLIPLF	Proportion of leaf tissue that is lipid (fraction). (/Plant/CROPGRO/GROW.for) Proportion of leaf tissue that is lipid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.025
PLIPST	Proportion of stem tissue that is lipid (fraction). (/Plant/CROPGRO/GROW.for) Proportion of stem tissue that is lipid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.020
PLIPRT	Proportion of root tissue that is lipid (fraction). (/Plant/CROPGRO/GROW.for) Proportion of root tissue that is lipid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.020
PLIPSH	Proportion of shell tissue that is lipid (fraction). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/IPPLNT.for) Proportion of shell tissue that is lipid (fraction)","IPPLNT, IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.020
PLIPNO	Proportion of nodule tissue that is lipid (fraction). (/Plant/CROPGRO/GROW.for) Proportion of nodule tissue that is lipid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.050
PLIGLF	Proportion of leaf tissue that is lignin (fraction). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/GROW.for) Proportion of leaf tissue that is lignin (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.070
PLIGST	Proportion of stem tissue that is lignin (fraction). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/GROW.for) Proportion of stem tissue that is lignin (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.070
PLIGRT	Proportion of root tissue that is lignin (fraction). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/GROW.for) Proportion of root tissue that is lignin (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.070
PLIGSH	Proportion of shell tissue that is lignin (fraction). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/GROW.for ; /Plant/CROPGRO/IPPLNT.for) Proportion of shell tissue that is lignin (fraction)","IPPLNT, IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.280
PLIGSD	Proportion of seed tissue that is lignin (fraction). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/GROW.for ;	.020

	/Plant/CROPGRO/IPPLNT.for ; /Plant/CROPGRO/PODS.for ; /Plant/CROPGRO/SDCOMP.for) Proportion of seed tissue that is lignin (fraction)","IPPLNT, IPDMND, PODCOMP, IPGROW, INC. (/Plant/CROPGRO/INCOMP.for)	
PLIGNO	Proportion of nodule tissue that is lignin (fraction). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/GROW.for) Proportion of nodule tissue that is lignin (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.070
POALF	Proportion of leaf tissue that is organic acid (fraction). (/Plant/CROPGRO/GROW.for) Proportion of leaf tissue that is organic acid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.050
POAST	Proportion of stem tissue that is organic acid (fraction). (/Plant/CROPGRO/GROW.for) Proportion of stem tissue that is organic acid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.050
POART	Proportion of root tissue that is organic acid (fraction). (/Plant/CROPGRO/GROW.for) Proportion of root tissue that is organic acid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.050
POASH	Proportion of shell tissue that is organic acid (fraction). (/Plant/CROPGRO/CROPGRO.for ; Plant/CROPGRO/GROW.for ; Plant/CROPGRO/IPPLNT.for) Proportion of shell tissue that is organic acid (fraction)","IPPLNT, IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.040
POASR	Proportion of storage tissue that is organic acid (fraction). Only for the forage model	
POASD	Proportion of seed tissue that is organic acid (fraction). (/Plant/CROPGRO/CROPGRO.for ; Plant/CROPGRO/DEMAND.for ; Plant/CROPGRO/GROW.for ; Plant/CROPGRO/IPPLNT.for ; Plant/CROPGRO/PODS.for ; Plant/CROPGRO/SDCOMP.for) Proportion of seed tissue that is organic acid (fraction)","IPPLNT, IPDMND, PODCOMP, IPGROW, INC. (/Plant/CROPGRO/INCOMP.for)	.040
POANO	Proportion of nodule tissue that is organic acid (fraction). (Plant/CROPGRO/GROW.for) Proportion of nodule tissue that is organic acid (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.050
PMINLF	Proportion of leaf tissue that is mineral (fraction). (Plant/CROPGRO/GROW.for) Proportion of leaf tissue that is mineral (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.094
PMINST	Proportion of stem tissue that is mineral (fraction). (Plant/CROPGRO/GROW.for) Proportion of stem tissue that is mineral (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.046
PMINRT	Proportion of root tissue that is mineral (fraction). (Plant/CROPGRO/GROW.for) Proportion of root tissue that is mineral (fraction)","IPGROW, INCOMP.	.057

	(/Plant/CROPGRO/INCOMP.for)	
PMINSH	Proportion of shell tissue that is mineral (fraction). (Plant/CROPGRO/CROPGRO.for ; Plant/CROPGRO/GROW.for ; Plant/CROPGRO/IPPLNT.for) Proportion of shell tissue that is mineral (fraction)","IPPLNT, IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.030
PMINSR	Proportion of storage tissue that is mineral (fraction). Only for the forage model	
PMINSD	Proportion of seed tissue that is mineral (fraction). (Plant/CROPGRO/CROPGRO.for ; Plant/CROPGRO/DEMAND.for ; Plant/CROPGRO/GROW.for ; Plant/CROPGRO/IPPLNT.for ; Plant/CROPGRO/PODS.for ; Plant/CROPGRO/SDCOMP.for) Proportion of seed tissue that is mineral (fraction)","IPPLNT, IPDMND, PODCOMP, IPGROW, INC. (/Plant/CROPGRO/INCOMP.for)	.025
PMINNO	Proportion of nodule tissue that is mineral (fraction). (Plant/CROPGRO/GROW.for ; Proportion of nodule tissue that is mineral (fraction)","IPGROW, INCOMP. (/Plant/CROPGRO/INCOMP.for)	.050

Note:

SDPROS, SDPROG - Standard protein concentration of seed, standard "growth" protein concentration of seed. Note: now only standard reference, changed by the cultivar-specific value in the ECO file. Also modified by C and N stresses as shown below.

!*SEED COMPOSITION VALUES

Table 4

Parameter	Parameter description	Soybean
LIPTB [1]	Temperature below which lipid composition is zero (°C). (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/SDCOMP.for) Base temperature of linear equation describing seed lipid concentration change in response to air temperature.	7.168
LIPOPT [1]	Temperature above which lipid composition is at a maximum (°C). (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/SDCOMP.for) Optimum temperature of linear equation describing seed lipid concentration change in response to air temperature.	23.65
SLOSUM*100 [2]	Slope of temperature vs. SUMTEM line (1/°C). (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/SDCOMP.for) Change (decrease) in sum of lipid and protein as air temperature increases	0.908
CARMIN	Minimum carbohydrate fraction. (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/SDCOMP.for) Minimum carbohydrate-cellulose concentration. Minimum amount of structure-cellulose-carbohydrates for seeds.	0.180

[1] LIPTB & LIPOPT - Solved for soybean by E. L. Piper from 20-years of Northern & and Southern Regional Variety Trials.

[2] Solved for soybean from Regional Variety Trials.

!*CARBON AND NITROGEN MINING PARAMETERS

Table 5

Parameter	Parameter description	Soybean
CMOBMX	Maximum C pool mobilization rate (g[CH2O] / m2 / d). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/IPPLNT.for; /Plant/CROPGRO/VEGGR.for) Maximum rate of mobilization of carbohydrate from vegetative tissues, fraction of available carbohydrate pool per day.	0.025
CADSTF	Proportion of CH2O reserves that are added to stalks (fraction). (/Plant/CROPGRO/VEGGR.for) Fraction of the carbohydrate reserves that are allocated to stem, (1-CADSTF) of carbohydrate reserves are stored in leaves.	0.75
CADPR1	Maximum fraction of stem growth after flowering that can be allocated to carbohydrate storage just before a full seed load is set. (fraction). (/Plant/CROPGRO/CROPGRO.for; /Plant/CROPGRO/IPPLNT.for) Carbohydrate accumulation continues after flowering (R1) until a full seed load is set.	.260
NMOBMX	Maximum Fraction of N which can be Mobilized per Day. (/Plant/CROPGRO/DEMAND.for) Maximum rate of mobilization of protein from vegetative tissues, during reproductive growth, fraction of available protein pool per day.	.090
NVSMOB	Relative rate of N mining during vegetative stage to that in reproductive stage. (/Plant/CROPGRO/DEMAND.for) Maximum rate of mobilization of carbohydrate from vegetative tissues, during vegetative growth, fraction of available protein pool per day	0.35
NRCVR	Fractional value between minimum and maximum tissue N values (0-1). (/Plant/CROPGRO/DEMAND.for) Check it. something to do with protein turnover in nodules. Not very important. Should be junked (KJB).	0.15
ALPHL	Fraction of new leaf growth that is mobile C (fraction). (/Plant/CROPGRO/GROW.for) Fraction of new leaf tissue growth that is available carbohydrate.	0.04
ALPHS	Fraction of new seed growth that is mobile C (fraction). (/Plant/CROPGRO/GROW.for) Fraction of new stem tissue growth that is available carbohydrate	0.08
ALPHR	Fraction of new root growth that is mobile C (fraction). (/Plant/CROPGRO/GROW.for) Fraction of new root tissue growth that is available carbohydrate	0.04
ALPHSH	Fraction of new shell growth that is mobile C (fraction). (/Plant/CROPGRO/GROW.for) Fraction of new shell tissue growth that is available carbohydrate.	0.08

!*NITROGEN FIXATION PARAMETERS

Table 6

Parameter	Parameter description	Soybean
SNACTM	Maximum activity of the nodules (g [nodule]). (/Plant/CROPGRO/NFIX.for) Specific nitrogenase activity of nodule tissues, g N fixed per g of nodule mass per day.	.045
NODRGM	Maximum nodule relative growth rate (g[nodule] / g[nodule] / d). (/Plant/CROPGRO/NFIX.for) Relative growth rate of nodules, g per g per day	.170
DWNODI	Initial nodule mass per plant (g[nodule] / plant). (/Plant/CROPGRO/NFIX.for) Initial nodule mass per plant, g per plant (Note, this really should be seed size specific. Need to change it.)	.014
TTFIX	Physiological days delay in nodule initiation (photo-thermal days / day). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/IPPLNT.for) Physiological days delay in nodule initiation (in practice, we have never had a non-zero value, i.e., need to start right away). Question, what is the starting point, emergence?	0.0
NDTHMX	Maximum relative death rate of nodules under flooded or dry conditions (g[nodule] / g[nodule] / d). (/Plant/CROPGRO/NFIX.for) Maximum nodule death rate, fraction per day.	0.07
CNODCR	C requirement for nodule respiration (g[C] / g[nodule] / d). (/Plant/CROPGRO/NFIX.for) Something about minimum arriving carbohydrate requirement per g. nodule mass per day, to prevent onset of senescence of nodule.	0.05
FNNGT(4), TYPNGT (Fig.5)	FNNGT(I) Critical temperature points for function to reduce Nodule growth rates when temperature is not optimal (°C). TYPNGT Type of function for temperature effects on Nodule growth rates. (/Plant/CROPGRO/NFIX.for) Temperature curve (LIN for linear 4-point function) describing relative nodule growth rate versus soil temperature (We don't know much about this one, and solve from field data, see Sexton et al., 1998).	7.00 22.0 35.0 44.0 LIN
FNFXT(4), TYPFXT (Fig.6)	FNFXT(I) Critical temperature points for function to reduce N fixation rate when temperature is not optimal (°C). TYPFXT Type of function for temperature effects on N fixation rates. (/Plant/CROPGRO/NFIX.for) Temperature curve (LIN for linear 4-point function) describing relative N-fixation rate versus soil temperature (We don't know much about this one either, and solved from field data, see Sexton et al., 1998).	5.00 20.0 35.0 44.0 LIN
FNFXD(4), TYPFXD (Fig.7)	FNFXD(I) Effects of soil water stress factor (TURFAC) on N fixation. TYPFXD Type of function for dry soil effects on N fixation. (/Plant/CROPGRO/NFIX.for) Relative effect of plant water deficit on N-fixation and nodule growth. Experience showed that using soil water in 5-40 cm soil zone did not give good results.	0.00 0.85 1.00 10.0 LIN
FNFXW(4), TYPFXW (Fig.8)	FNFXW(I) Critical soil volumetric water contents for reducing N fixation when the soil is too wet (flooded) (cm3/cm3). TYPFXW Type of function for wet soil effects on N fixation. (/Plant/CROPGRO/NFIX.for) Relative effect of excess soil water (between DUL and SAT) in 5-40 cm soil zone on N-fixation and nodule growth. Check this one.	-.02 .001 1.00 2.00 LIN
FNFXA(4), TYPFXA	FNFXA(I) Critical points in development from first seed to physiological maturity for reducing N fixation due to canopy physiological age.	0.00 0.10 1.00 0.00 INL

(Fig.9) TYPFXA Type of function for canopy age effects on N fixation. (/Plant/CROPGRO/NFIX.for)
Relative aging effect on N-fixation after beginning seed. check this one

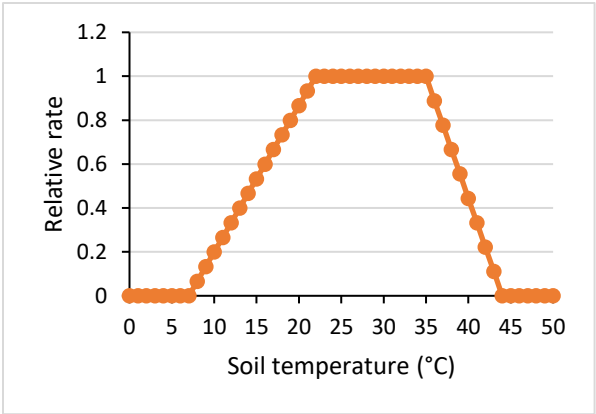

Figure 5. Temperature curve (LIN for linear 4-point function) describing relative nodule growth rate versus soil temperature.

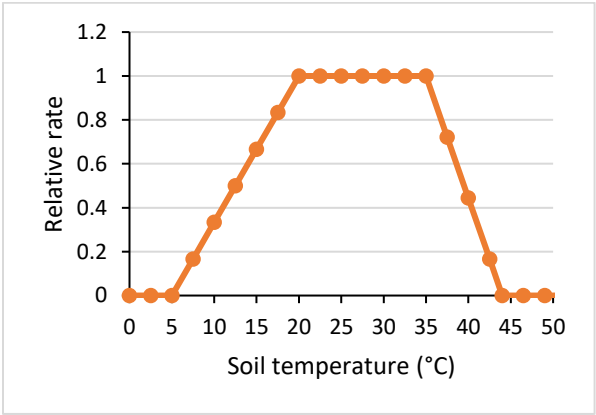

Figure 6. Temperature curve (LIN for linear 4-point function) describing relative N-fixation rate versus soil temperature

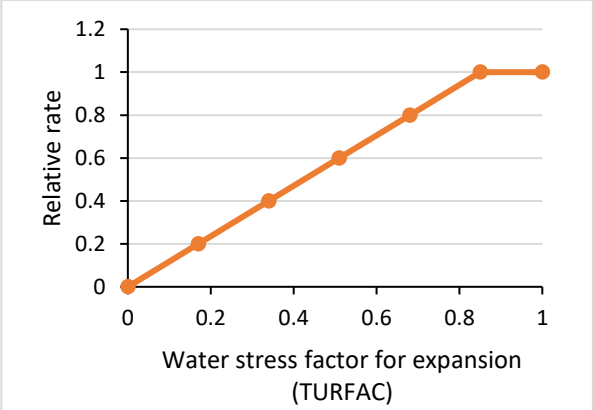

Figure 7. Relative effect of plant water deficit on N-fixation and nodule growth.

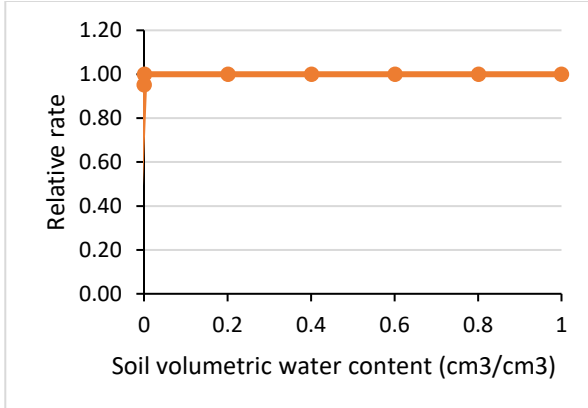

Figure 8. Relative effect of excess soil water (between DUL and SAT) in 5-40 cm soil zone on N-fixation and nodule growth.

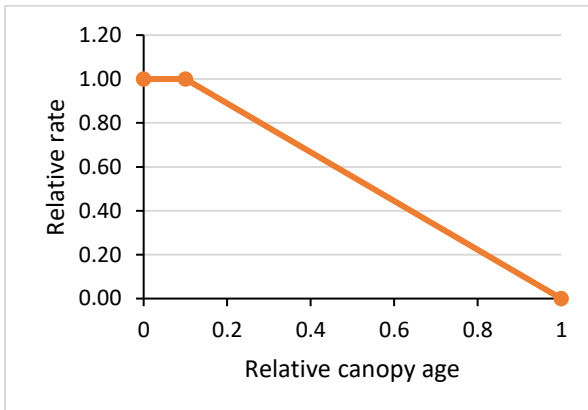

Figure 9. Relative aging effect on N-fixation after beginning seed

!*VEGETATIVE PARTITIONING PARAMETERS

Table 7

Parameter	Parameter description	Soybean
XLEAF (Fig.10) YLEAF (Fig.10)	XLEAF(I) V-stage at which partitioning to leaves is YLEAF(I). (leaf nodes). YLEAF(I) Partitioning fraction to leaves at V-stage XLEAF(I) (g[leaf] / g[veg. plant]). (/Plant/CROPGRO/DEMAND.for) Describes dry matter partitioning among vegetative tissue only, as a function of vegetative stage, to leaf, stem, and root. The total must equal 1.0	0.0 1.5 3.3 5.0 7.8 10.5 30.0 40.0 0.41 0.42 0.42 0.41 0.36 0.32 0.31 0.31
YSTEM (Fig.10)	YSTEM(I) Partitioning factor for stem growth at V-stage XSTEM(I) (g[stem] / g[veg. plant]) (/Plant/CROPGRO/DEMAND.for) Describes dry matter partitioning among vegetative tissue only, as a function of vegetative stage, to leaf, stem, and root. The total must equal 1.0	0.09 0.13 0.21 0.29 0.37 0.49 0.49 0.49
WTFSD	Relative weight of seed compared to maximum (fraction). (/Plant/CROPGRO/GROW.for) Fraction of initial seed dry weight that is plant mass at emergence. Adjusts for different seed size.	0.55
PORPT	Ratio of petiole to leaf weight. (/Plant/CROPGRO/SENES.for)	0.58

	When leaf mass senesces, stem mass also is abscised at this fraction of the senesced leaf mass. This is a stand-in for petiole mass (not a real state variable in the model). It can be thought of the ratio of petiole mass to leaf mass. Needs to be small for plant that does not lose petioles.	
FRSTMF	Fraction of daily dry weight increase in vegetative plant parts which goes to stems after the day on which the maximum number of V-stages occurs (NDVSTG). (g[stem] / g[veg]). (/Plant/CROPGRO/DEMAND.for) The fraction of vegetative dry matter growth allocated to stem at the "final" stage, NDLEAF, when leaf expansion is totally finished.	0.55
FRLFF	Fraction of daily increase in vegetative weight which goes to leaves after the day on which the maximum number of V-stages occurs (NDVSTG). (g[leaf] / g[veg]). (/Plant/CROPGRO/DEMAND.for) The fraction of vegetative dry matter growth allocated to leaf at the "final" stage, NDLEAF, when leaf expansion is totally finished.	0.24
ATOP	Maximum fraction change in partitioning from top growth to roots if severe water or nitrogen stresses occur. (/Plant/CROPGRO/VEGGR.for) The maximum fraction of shoot dry matter that is allowed to be diverted to root growth under maximum water deficit (SWFAC = 0.0) in response to water deficit.	1.00
FRCNOD	Fraction of new root dry matter allocation that is diverted to nodule growth. (/Plant/CROPGRO/IPPLNT.for "Bypass" flow to nodule growth, the fraction of new root dry matter allocation that is diverted to nodule growth, primarily acts during early vegetative growth. Shown necessary by modeling experience, to prevent lags in nodulation and N-fixation.	0.05

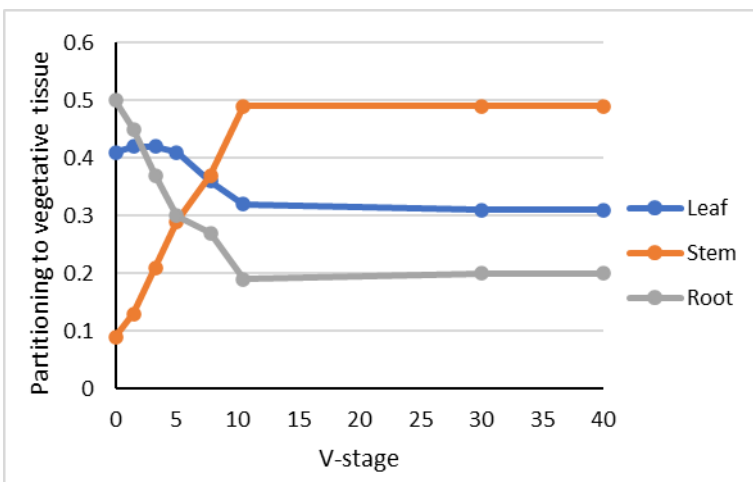

Figure 10. Dry matter partitioning among vegetative tissue only, as a function of vegetative stage, to leaf (blue line), stem (orange line), and root (gray line). The total must equal 1.0

!*LEAF GROWTH PARAMETERS

Table 8

Parameter	Parameter description	Soybean

FINREF	Specific leaf area (SLA) of leaves of standard crop cultivar when plants emerge (cm2[leaf] / g[leaf]). (/Plant/CROPGRO/DEMAND.for) The specific leaf area (cm2/g) of leaves at plant emergence, scaled via SLAVAR though.	180.
SLAREF	Specific leaf area (SLA) for new leaves during peak vegetative growth for the standard cultivar. (cm2/g). (/Plant/CROPGRO/DEMAND.for) The specific leaf area (cm2/g) of the standard reference cultivar at peak early vegetative phase, under optimum temperature, water, and light.	350.
SIZREF	The size of a normal upper node leaf (nodes 8 – 10) of standard cultivar. (cm2 / leaf). (/Plant/CROPGRO/DEMAND.for) Supposedly leaf area per trifoliate (tetrafoliate) leaf for the leaf at the 5th node position. Actually, we use it to modify the rate of leaf area expansion for the first VSSINK leaf nodes, to limit rate of leaf area expansion until VSSINK	171.4
VSSINK	Vegetative stage beyond which sink-limited leaf area expansion can no longer limit photosynthesis or leaf area growth. (/Plant/CROPGRO/DEMAND.for) The vegetative growth stage though which leaf area expansion and photosynthesis can be sink-limited. After this vegetative growth stage, growth is totally source-driven.	5.0
EVMODC	Modifier of rate of vegetative node appearance for the first few nodes, primarily used for peanut. (/Plant/CROPGRO/Ipphenol.for ; /Plant/CROPGRO/PHENOL.for) A modifier of rate of vegetative node appearance for the first few nodes, primarily used for peanut. Not particularly important.	0.0
SLAMAX, SLAMIN, SLAPAR	SLAMAX – The maximum specific leaf area (SLA) for new leaves when grown under low (nearly zero) radiation but optimum water and temperature for the standard cultivar. (cm2 / g). SLAMIN – The minimum specific leaf area (SLA) for new leaves when grown under infinitely high radiation, optimum water and temperature for the standard cultivar. (cm2 / g). SLAPAR – Coefficient in exponential equation to reduce SLA as PAR increases (leaf curvature). (/Plant/CROPGRO/DEMAND.for) These three describe SLA response of today's new leaf area to solar irradiance: SLAMAX is the (thinnest) leaves can be under low light, whereas SLAMIN is the (thickest) leaves can be under high light, and SLAPAR describes the curvature. The function and curvature come from Cox et al. for peanut response to light in a phytotron. SLAMAX and SLAMIN set the SLA of a given species. SLAVAR or SLAREF should not be used for that.	950. 250.0 -.048
TURSLA	Water stress effects on leaf area expansion. (/Plant/CROPGRO/DEMAND.for) Modifier of water stress (TURFAC) effect on leaf area expansion. Minor.	1.50
XVGROW(1-6) (Fig.11)	XVGROW(I) V-stage at which maximum leaf area growth per plant since emergence is YVGROW(I). (# leaf nodes). (/Plant/CROPGRO/DEMAND.for) Lookup table of V stage values.	0.0 1.0 2.0 3.0 4.0 5.0
YVGROW(I)	Maximum leaf area grown per plant at V-stage XVGROW(I) (cm2 / plant). (/Plant/CROPGRO/DEMAND.for) Is this being used at all?	??
YVREF(1-6) (Fig.11)	YVREF(I) Maximum leaf area grown per plant at V-stage XVGROW(I), for reference cultivar. (cm2 / plant). (/Plant/CROPGRO/DEMAND.for)	0.0 20.0 55.0 110.0 200.0 320.0

	Respective maximum leaf area (cm2/plant) at corresponding V stage, part of possible limiting leaf area expansion for first VSSINK nodes.	
XSLATM(1-5) (Fig.12)	XSLATM(I) Temperature values for function that reduces specific leaf area (SLA) (°C). (/Plant/CROPGRO/DEMAND.for)	-50.0 00.0 12.0 22.0 60.0
YSLATM(1-5) (Fig.12)	Relative temperature effect on specific leaf area of newly-formed leaves YSLATM(I) Array which describes the effect of temperature on specific leaf area. (/Plant/CROPGRO/DEMAND.for)	0.25 0.25 0.25 1.00 1.0

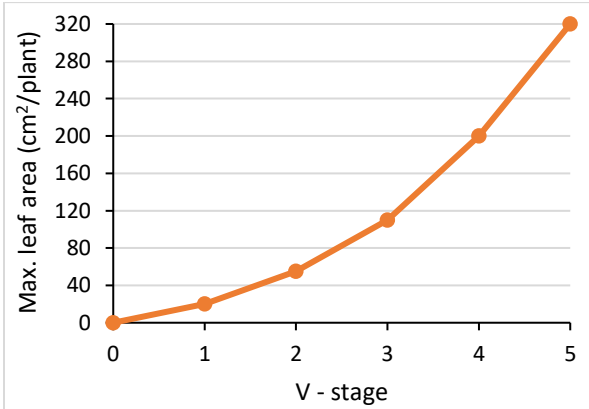

Figure 11. XVGROW(1-6) vs. YVREF(1-6) - Respective maximum leaf area (cm2/plant) at corresponding V stage, part of possible limiting leaf area expansion for first VSSINK nodes.

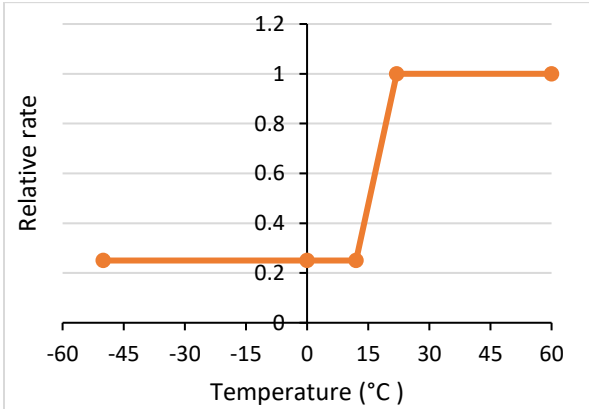

Figure 12. XSLATM(1-5), YSLATM(1-5) - Relative temperature effect on specific leaf area (SLA) of newly-formed leaves.

!*LEAF SENESCENCE FACTORS

Table 9

Parameter	Parameter description	Soybean
SENRTE	Factor by which protein mined from leaves each day is multiplied to determine LEAF senescence. (/Plant/CROPGRO/SENES.for) grams of leaf mass (already exhausted of available leaf protein) that is lost per gram of protein mobilized.	0.80
SENRT2	Factor by which leaf weight is multiplied to determine senescence each day after NR7 (g(leaf) / g(protein loss)) (/Plant/CROPGRO/SENES.for) Rate of leaf abscission after physiological maturity (R7), fraction per day.	0.20

SENDAY	Maximum fraction of existing leaf weight which can be senesced on day N as a function of severe water stress 4 days earlier. (/Plant/CROPGRO/SENES.for) Maximum rate of leaf abscission, fraction per day, in response to water deficit.	0.06
FREEZ1	Temperature below which plant loses all leaves, but development continues (°C). (/Plant/CROPGRO/CROPGRO.for ; Plant/CROPGRO/FREEZE.for, not listed at the bottom ; Plant/CROPGRO/IPPLNT.for). Temperature below which leaves are killed, but stems, pods, and seeds remain alive, development to maturity proceeds, and seeds can grow on mobilized reserves.	-2.22
FREEZ2	Temperature below which plant growth stops completely. (°C). (/Plant/CROPGRO/CROPGRO.for ; Plant/CROPGRO/FREEZE.for ; Plant/CROPGRO/IPPLNT.for). Temperature below which development and crop growth simulation stops.	-5.00
XSTAGE (Fig.13)	V-stage at which SENPOR(I) fraction of cumulative leaf growth will have been senesced if no water stress occurred. (/Plant/CROPGRO/SENES.for)	0.0 5.0 14.0 30.0
SENPOR (Fig.13)	Proportion of leaf weight grown which will have been senesced by a given V- stage (XSTAGE(I)) if no water stress has occurred prior to this V-stage (XSTAGE(I)) – normal vegetative senescence does not occur if prior water stress has already reduced leaf. (/Plant/CROPGRO/SENES.for) SENPOR vs. XSTAGE - The fraction leaf abscission that normally occurs by a given V stage	0.0 0.0 0.12 0.12
XSENMX (Fig.14)	V-stage at which maximum fraction of cumulative leaf growth vulnerable to loss due to water stress is SENMAX(I). (# leaf nodes). (/Plant/CROPGRO/SENES.for)	3.0 5.0 10.0 30.0
SENMAX (Fig.14)	Maximum proportion of total leaf weight as a function of V-stage (XSENMX(I)) which can be senesced due to water stress. (/Plant/CROPGRO/SENES.for) SENMAX vs. XSENMX - The maximum allowable leaf abscission due to water stress that is allowed to occur by a given V stage.	0.0 0.2 0.6 0.6

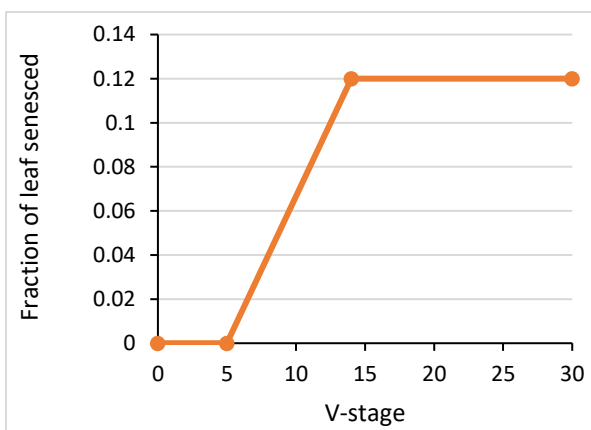

Figure 13. SENPOR vs. XSTAGE - The fraction leaf abscission that normally occurs by a given V stage

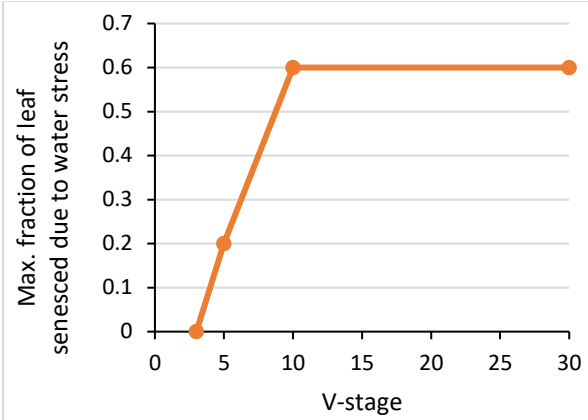

Figure 14. SENMAX vs. XSENMX - The maximum allowable leaf abscission due to water stress that is allowed to occur by a given V stage.

!*ROOT PARAMETERS

Table 10

Parameter	Parameter description	Soybean
RTDEPI	Depth of roots on day of plant emergence. (cm) (/Plant/CROPGRO/ROOTS.for) The initial root depth at emergence, cm.	25.0
RFAC1	Root length per unit root weight. (cm/g). (/Plant/CROPGRO/ROOTS.for) Root length per unit mass (cm per g).	7500.
RTSEN	Fraction of existing root length which can be senesced per physiological day. (fraction / ptd). (/Plant/CROPGRO/ROOTS.for) Root senescence, fraction per physiological day	0.020
RLDSM	Minimum root length density in a given layer, below which drought-induced senescence is not allowed (cm [root]/ cm3 [soil]). (/Plant/CROPGRO/ROOTS.for)	0.1
RTSDF	Maximum fraction of root length senesced in a given layer per physiological day when water content in a given layer falls below 25 % of extractable soil water. (/Plant/CROPGRO/ROOTS.for)	.015
RWUEP1	Threshold for reducing leaf expansion compared w/ ratio of TRWU/EP1 (total potential daily root water uptake/ actual transpiration). (/Plant/CROPGRO/CROPGRO.for) This is the ratio of evaporative demand:root water uptake at which TURFAC is 1.00, and declines below that ratio.	1.50
RWUMX	Maximum water uptake per unit root length, constrained by soil water (cm3[water] / cm [root]). (/Plant/CROPGRO/CROPGRO.for)	0.04
XRTFAC (Fig.15)	XRTFAC(I) V-stage at which rate of increase in root depth per physiological day is YRTFAC(I). (# leaf nodes). (/Plant/CROPGRO/ROOTS.for)	0.0 2.50 3.0 2.50
YRTFAC (Fig.15)	Rate of increase in root depth per degree day at V-stage XRTFAC(I). (cm / (physiol. day)). (/Plant/CROPGRO/ROOTS.for) YRTFAC vs. XRTFAC - Maximum rate of root depth progression, cm per physiological day	6.0 2.50 30.0 2.50
RTNO3	Nitrate uptake per unit root length (mg N / cm). (/Plant/CROPGRO/NUPTAK.for)	0.006

RTNH4	Maximum uptake of NO3 per unit root length. Ammonium uptake per unit root length (mg N / cm). (/Plant/CROPGRO/NUPTAK.for) Maximum uptake of NH4 per unit root length.	0.006

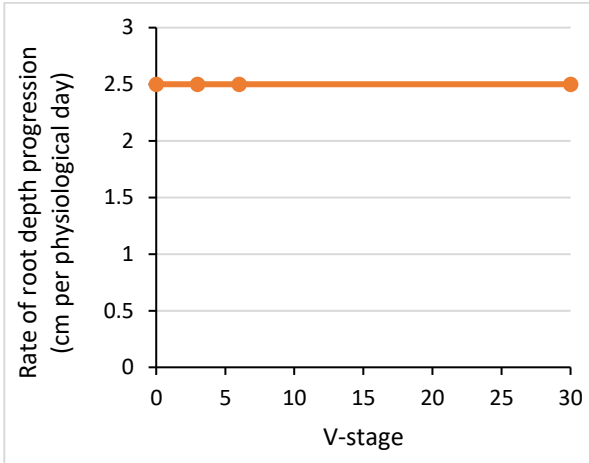

Figure 15. YRTFAC vs. XRTFAC - Maximum rate of root depth progression, cm per physiological day.

!*SEED AND SHELL GROWTH PARAMETERS

Table 11

Parameter	Parameter description	Soybean
SETMAX	Fraction of pod and seed number to set in a given cohort if the supply to demand ratio for that cohort is less than 1.0. (/Plant/CROPGRO/PODS.for) Fraction of pod and seed number to set in a given cohort if the supply to demand ratio for that cohort is less than 1.0. This is a multiplier that acts only when the last few pods are being added. For that reason, it is not very important and has minor effect.	0.60
SRMAX	Maximum fraction change in seed growth rate for long day lengths. (/Plant/CROPGRO/DEMAND.for) The fraction decrease in potential single seed growth rate, under completely non optimum daylength	0.3
RFLWAB	relative rate of flower abortion per day because daylength is not optimum. The flowers that survive "NR2". Relative rate of flower abortion per day because daylength not optimum (/Plant/CROPGRO/PODS.for, description listed twice) Don't know this. May not function. Flower abortion??	0.00
XMPAGE	Maximum physiological age of a fruit beyond which shell thickening and seed growth end. (p-t-d). (/Plant/CROPGRO/PODS.for) Not used, I think. Maximum age of a cohort beyond which seed growth stops. It might work.	100.
DSWBAR	Average depth of soil over which soil water content affects flower, pod addition (cm). (/Plant/CROPGRO/PODS.for) The depth of the fruiting zone for peanut, in which fraction available soil water affects peanut fruiting.	15.0

XFRMAX	Maximum increase in partitioning to fruits induced under water stress, assuming no problem in pod setting. (/Plant/CROPGRO/DEMAND.for) Degree of increase in partitioning to pods, rather than vegetative growth, when water deficit occurs during pod growth.	0.0
SHLAG	Shell (peg) growth rate during its initial slow growth phase after beginning pegging (R2) as a fraction of shell growth rate (SHVAR) during its rapid growth phase. (/Plant/CROPGRO/DEMAND.for ; /Plant/CROPGRO/PODS.for) Slow growth phase (lag) in shell growth (used primarily for peanut).	0.0
FNPDT(1-4), TYPPDT (Fig.16)	FNPDT(I) Critical values of temperature for function to reduce pod addition and seed setting rates under non-optimal temperatures (°C). TYPPDT Type of function for temperature effects on pod and seed addition rates. (/Plant/CROPGRO/PODS.for) Relative temperature effect (2-sided parabolic shape, two middle points are at optimum) that affects pod addition rate. Taken from several data sources. Recent evidence indicates that soybean cultivars vary in this sensitivity of pod set to cool temperature. Will use ecotype trait in next version to modify this curve. Recent evidence shows opt temp goes higher.	14.0 21.0 26.5 40.0 QDR
FNSDT(1-4), TYPSDT (Fig.17)	FNSDT(I) Temperature values which describe function for modifying seed growth rate with temperature (°C). TYPSDT Curve type for temperature factor calculations (for use in function subroutine CURV). (/Plant/CROPGRO/DEMAND.for) Relative temperature effect (2-sided parabolic shape, two middle points are at optimum) that affect single seed growth rate. Function is from Egli and Wardlaw, above end confirmed in Florida SPAR experiments.	6.0 21.0 23.5 41.0 QDR
XXFTEM(1-6) (Fig.18)	XXFTEM(I) Array of temperature values in table lookup describing effect of temperature on partitioning to pods (YXFTEM = 0 TO 1). (°C). (/Plant/CROPGRO/DEMAND.for)	0.00 5.00 20.0 35.0 45.0 60.0
YXFTEM(1-6) (Fig.18)	YXFTEM(I) Array describing the relative partitioning to pods (0 to 1 effect on XFRUIT) as temperature increases. (/Plant/CROPGRO/DEMAND.for) YXFTEM(1-6) vs. XXFTEM(1-6) - Describes a 6-point curve of relative partitioning versus temperature. Good evidence for temperature effects at high end, but KJB thinks that the YXFTEM should be 1.0 below 20. Let slower phenology (from cool temperature) or slower numbers of pods set, as the primary reason to decrease yield at lower end.	1.00 1.00 1.00 1.00 0.00 0.00
XSWFAC(1-4) (Fig.19)	XSWFAC(I) Array of values describing the ratio of soil-root water supply to transpirative demand, at which the rate of pod addition is reduced by a given proportion (YSWFAC). (/Plant/CROPGRO/PODS.for)	0.00 0.50 1.00 1.00
YSWFAC(1-4) (Fig.19)	YSWFAC(I) Array describing the relative (0 to 1) rate of pod addition (pegging) which occurs at a given ratio of soil-root water supply to transpirative demand (XSWFAC). (/Plant/CROPGRO/PODS.for) YSWFAC(1-4) vs. XSWFAC(1-4) - Describes soil water deficit effect on pod addition, in addition to effects on assimilate supply. I'm not sure this function is needed or advisable.	0.00 1.00 1.00 1.00
XSWBAR(1-5) (Fig.20)	XSWBAR(I) Array of available soil water content in the pegging zone, at which rate of pod addition (pegging) is affected in a given proportion (YSWBAR). (cm3/cm3). (/Plant/CROPGRO/PODS.for)	0.00 0.01 0.25 1.00 1.00
YSWBAR(1-5) (Fig.20)	YSWBAR(I) Array describing the relative (0 to 1) rate of pod addition (pegging) which occurs at a given fraction available soil water content in the pegging zone.	1.00 1.00 1.00 1.00 1.00

(/Plant/CROPGRO/PODS.for)
YSWBAR(1-5) vs. XSWBAR(1-5) - Relative response of peanut fruit pegging in response to soil water available only in the top SWBAR of soil. Function should not have any effect (should be 1.0s) on soybean or common bean.

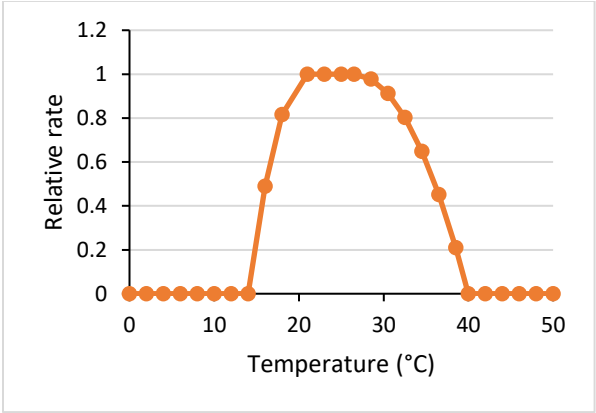

Figure 16. Relative temperature effect (2-sided parabolic shape, two middle points are at optimum) that affects pod addition rate. Taken from several data sources. Recent evidence indicates that soybean cultivars vary in this sensitivity of pod set to cool temperature. Will use ecotype trait in next version to modify this curve. Recent evidence shows opt temp goes higher.

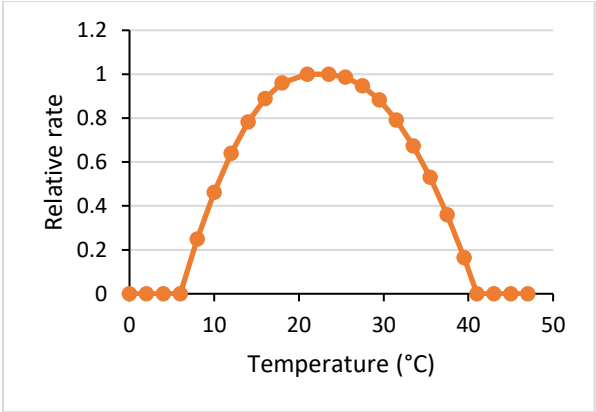

Figure 17. Relative temperature effect (2-sided parabolic shape, two middle points are at optimum) that affect single seed growth rate. Function is from Egli and Wardlaw, above end confirmed in Florida SPAR experiments.

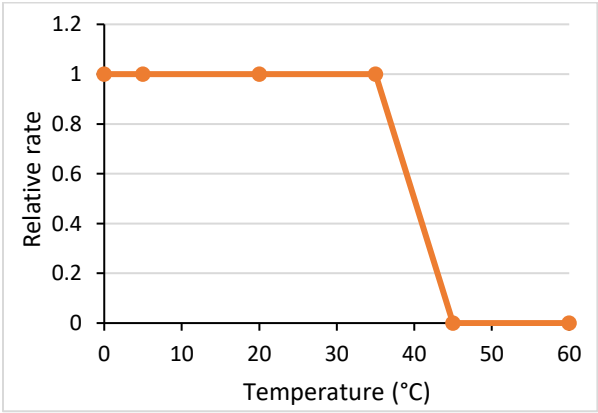

Figure 18. Describes a 6-point curve of relative partitioning versus temperature. Good evidence for temperature effects at high end, but KJB thinks that the YXFTEM should be 1.0 below 20. Let slower phenology (from cool temperature) or slower numbers of pods set, as the primary reason to decrease yield at lower end.

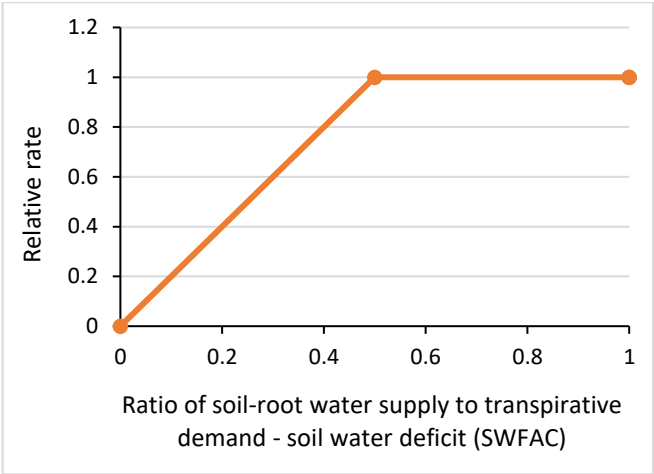

Figure 19. Describes soil water deficit effect on pod addition, in addition to effects on assimilate supply.

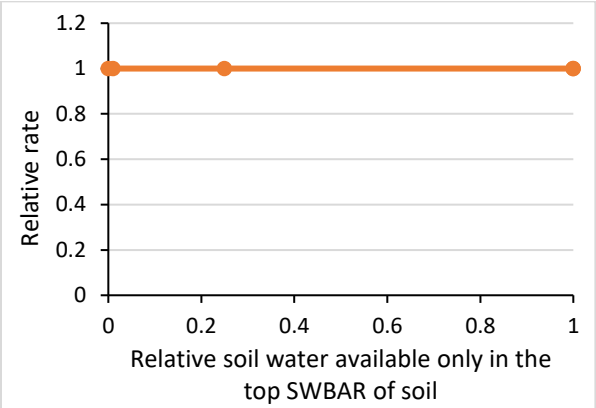

Figure 20. Relative response of peanut fruit pegging in response to relative soil water available only in the top SWBAR of soil. Function should not have any effect (should be 1.0s) on soybean or common bean.

!*POD LOSS PARAMETERS

Table 12

Parameter	Parameter description	Soybean
DETACH	Switch to determine if pod detachment will be simulated (Y or N). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/IPPLNT.for)	N
DWC	Threshold number of days without carbon to trigger pod detachment (days). (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/PODDET.for)	6.0
PR1DET	Threshold for comparison of ratio of leaf mass to pod mass (RLMPM). (/Plant/CROPGRO/PODDET.for)	.3961

PR2DET	PR2DET Threshold for comparison of slope of leaf mass (for defoliation). (/Plant/CROPGRO/PODDET.for)	-.865
XP1DET	Coefficient which defines pod detachment equation. (/Plant/CROPGRO/PODDET.for)	1.00
XP2DET	Coefficient which defines pod detachment equation. (/Plant/CROPGRO/PODDET.for)	0.00

Note: Without explanation, the above are pod loss functions that can be turned on (by flagging Y) to cause pod losses for peanut. The calibrations are for Florunner peanut by W. D. Batchelor. Keep it flagged "N", unless you know what you are doing.

!*PHENOLOGY PARAMETERS

Table 13

Parameter	**Parameter description**	**Soybean**
1 Vegetative development		
TB	Base temp (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) Base temperature	7.0
TO1	1st optimum (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) First optimum	28.0
TO2	2nd optimum (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) Second optimum	35.0
TM	Max temp. (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) Maximum temperature	45.0
2 Early reproductive development		
TB	Base temp (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) Base temperature	6.0
TO1	1st optimum (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) First optimum	26.0
TO2	2nd optimum (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) Second optimum	30.0
TM	Max temp. (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed) Maximum temperature	45.0

3 Late reproductive development		
TB	Base temp (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed)	-15.0
	Base temperature	
TO1	1st optimum (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed)	26.0
	First optimum	
TO2	2nd optimum (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed)	34.0
	Second optimum	
TM	Max temp. (°C) (/Plant/CROPGRO/PODDET.for ; /Plant/CROPGRO/Ipphenol.for, not listed ; /Plant/CROPGRO/PHENOL.for, not listed)	45.0
	Maximum temperature	

Note: These are 4-point look-up functions of base temperature (TB), first optimum (TO1), second optimum (TO2), and maximum temperature (TM). Relative rate is zero at TB and TM, and optimum (1.0) at TO1 and TO2. Possible to use linear (LIN) or parabolic (QDR) between these points. The Flag 1, 2, or 3 is a flag for temperature function affecting vegetative (V staging) only, early reproductive, and late reproductive development. *Bulbing (with available photoperiod) is stimulated by higher temperatures

See the use of 1, 2, or 3 in the phenology development "road-map" below.

!FOLLOWING LINE: STAGE; REF STAGE; PHOTOPERIOD FUNCTION; TEMPERATURE FUNCT;
!POINTER TO VEGD(1) OR REPDA(2) OR REPDB(3) TEMP SENS; SENS TO WATER;N; AND P

```
1    1 NON    LIN 1 -0.20 0.00 0.00 PLANT(STG 1) TO EMERG(STG 2) PHASE
2    2 NON    LIN 1 -0.20 0.00 0.00 EMERG(STG 2) TO V1(STG 3) PHASE
3    2 NON    LIN 1 -0.40 0.00 0.00 EMERG(STG 2) TO END JV(STG 4) PHASE
4    4   INL   LIN 2 -0.40 0.00 0.00 END JV(STG 4) TO FL IND(STG 5) PHASE
5    5   INL   LIN 2 -0.40 0.00 0.00 FL IND(STG 5) TO 1ST FL(STG 6) PHASE
6    6   INL   LIN 2 -0.40 0.00 0.00 1ST FL(STG 6) TO 1ST PEG(STG 7) PHASE
7    6   INL   LIN 2 -0.40 0.00 0.00 1ST FL(STG 6) TO 1ST POD(STG 8) PHASE
8    6   INL   LIN 2 -0.40 0.00 0.00 1ST FL(STG 6) TO 1ST SD(STG 9) PHASE
9    9   INL   LIN 3  0.70 0.40 0.00 1ST SD(STG 9) TO LST SD(STG 10) PHASE
10   9   INL   LIN 3 0.70 0.40 0.00 1ST SD(STG 9) TO PH MAT(STG 11) PHASE
11 11 NON NON 1 0.00 0.00 0.00 PH MAT(STG 11) TO H-MAT(STG 12) PHASE
12   6 INL      LIN 2 -0.60 0.00 0.00 1ST FL(STG 6) TO LST VST(STG 13) PHASE
13   6 INL      LIN 2 -0.90 0.00 0.00 1ST FL(STG 6) TO LST LF(STG 14) PHASE
```

There are 13 life cycle phases (STAGE). The second value (REF STAGE) indicates the stage at which the new stage accumulator can start. The third parameter indicates if there is no photoperiod sensitivity (NON) or photoperiod sensitivity (INL = CSDL with PP-SEN slope). The fourth parameter indicates the whether the temperature function is to use linear (LIN) or quadratic (QDR) or sine (SIN) function and the fifth parameter (1, 2, or 3) indicates which temperature function to use. The 6th, 7th, and 8th parameters indicate whether development in that phase is sensitive to water deficit, N deficit, or P deficit (latter does not exist yet). The phases on the right verbally describe the life cycle phase.

!*CANOPY HEIGHT AND WIDTH GROWTH PARAMETERS
! VSTAGE, FOLLOWED BY INTERNODE LENGTH PER NODE, THEN CANOPY WIDTH PER NODE

Table 14

Parameter	Parameter description	Soybean
XVSHT(1-10) (Fig.21)	Node number on main stem for use in computing height and width growth rates. (/Plant/CROPGRO/CANOPY.for)	0.00 1.00 4.00 6.00 8.00 10.00 14.00 16.00 20.00 40.00
YVSHT(1-10) (Fig.21A)	Length of internode (m) Vs position on the main stem defined by XVSHT (m/node). (/Plant/CROPGRO/CANOPY.for) YVSHT(1-10) and YVSHH(1-10) vs. XVSHT(1-10) - describe the internode length (meters per internode) and canopy width increase per internode, as a function of plant Vegetative node stage. Early internodes are shorter, middle ones longer, and some crops decrease length of later internodes.	.0300 .0530 .0630 .0660 .0690 .0660 .0620 .0510 .0340 .0060
YVSWH(1-10) (Fig.21B)	Increase in canopy width per node developed on the main stem (m/node). (/Plant/CROPGRO/CANOPY.for) YVSHT(1-10) and YVSHH(1-10) vs. XVSHT(1-10) - describe the internode length (meters per internode) and canopy width increase per internode, as a function of plant Vegetative node stage. Early internodes are shorter, middle ones longer, and some crops decrease length of later internodes.	.0300 .0510 .0620 .0640 .0660 .0630 .0590 .0460 .0250 .0010
XHWTEM(1-5) (Fig.22)	Temperatures in a table look-up function for modifying height and width growth rates (°C). (/Plant/CROPGRO/CANOPY.for)	-50.0 00.0 15.0 26.0 60.0
YHWTEM(1-5) (Fig.22)	YHWTEM(I) Relative (0-1) expansion in height and width with temperatures given in XHWTEM. (/Plant/CROPGRO/CANOPY.for) YHWTEM(1-5) vs. XHWTEM(1-5) - Describes relative effect of temperature on internode length.	0.40 0.40 0.50 1.00 1.00

XHWPAR(1-8) (Fig.23)	XHWPAR(I) PAR values for table look-up for modifying height and width growth rate, particularly to allow etiliolation at low PAR values (mol/day). (/Plant/CROPGRO/CANOPY.for)	0.00 5.00 7.50 10.00 15.00 20.00 30.00 80.00
YHWPAR(1-8) (Fig.23)	YHWPAR(I) Relative increase in height and width growth rates with low PAR as given in XHWPAR. (/Plant/CROPGRO/CANOPY.for)	4.00 2.00 1.50 1.25 1.05 1.00 1.00 1.00
	YHWPAR(1-8) vs. XHWPAR(1-8) - Describes relative internode elongation versus photosynthetic photon flux density per day (per mole).	

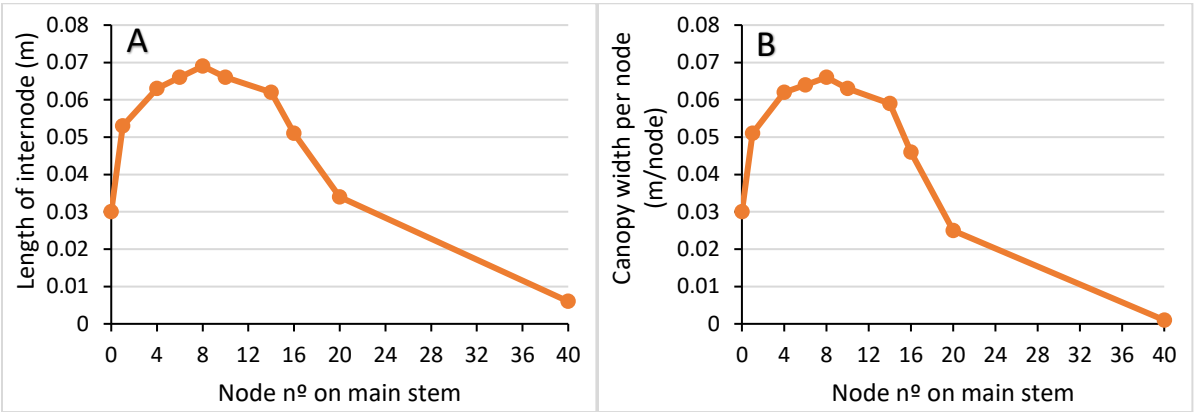

Figure 21. YVSHT(1-10) and YVSHH(1-10) vs. XVSHT(1-10) - describe the internode length (meters per internode) (A) and canopy width increase per internode (B), as a function of plant Vegetative node stage. Early internodes are shorter, middle ones longer, and some crops decrease length of later internodes.

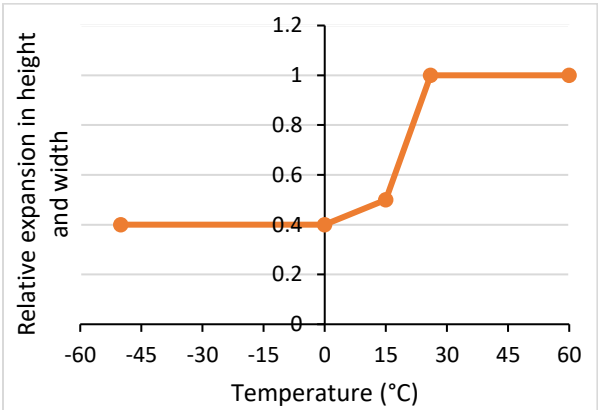

Figure 22. YHWTEM(1-5) vs. XHWTEM(1-5) - Describes relative effect of temperature on internode length.

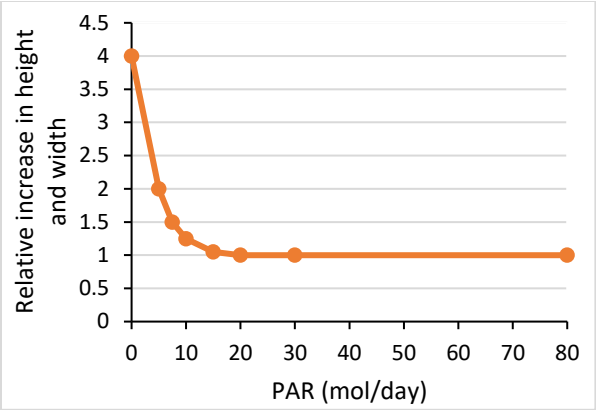

Figure 23. YHWPAR(1-8) vs. XHWPAR(1-8) - Describes relative internode elongation versus photosynthetic photon flux density per day (per mole).

!*EVAPOTRANSPIRATION

Table 15

Parameter	Parameter description	Soybean
KEP	Energy extinction coefficient for partitioning EO to EP (/Plant/CROPGRO/CROPGRO.for ; /Plant/CROPGRO/IPPLNT.for – not listed at the bottom) Extinction coefficient for solar radiation for partitioning of potential ET to T. Formerly had KCAN+0.15. That was too high, as shown by experiments by Sau and Boote. Decreasing this from 0.85 to 0.70 for V4.0, improved early season (created less water stress). It should be similar to KCAN which was decreased from 0.70 to 0.68 for V4.0. Keep separate for calibration purposes for now.	0.70
EORATIO	Ratio of increase in potential evapotranspiration with increase in LAI (up to LAI=6.0) for use with FAO-56 Penman reference potential evapotranspiration. (/Plant/CROPGRO/CROPGRO.for) To be used only for the new FAO-56 reference method of ET computation to increase ET above reference method as a function of increasing LAI. Suggest a value of about 1.1 for soybean and faba bean. This increases potential ET by ratio 1.1 at LAI of 6, compared to the reference (short grass with LAI of 2.88). Approach also used by CROPSYS.	1.0